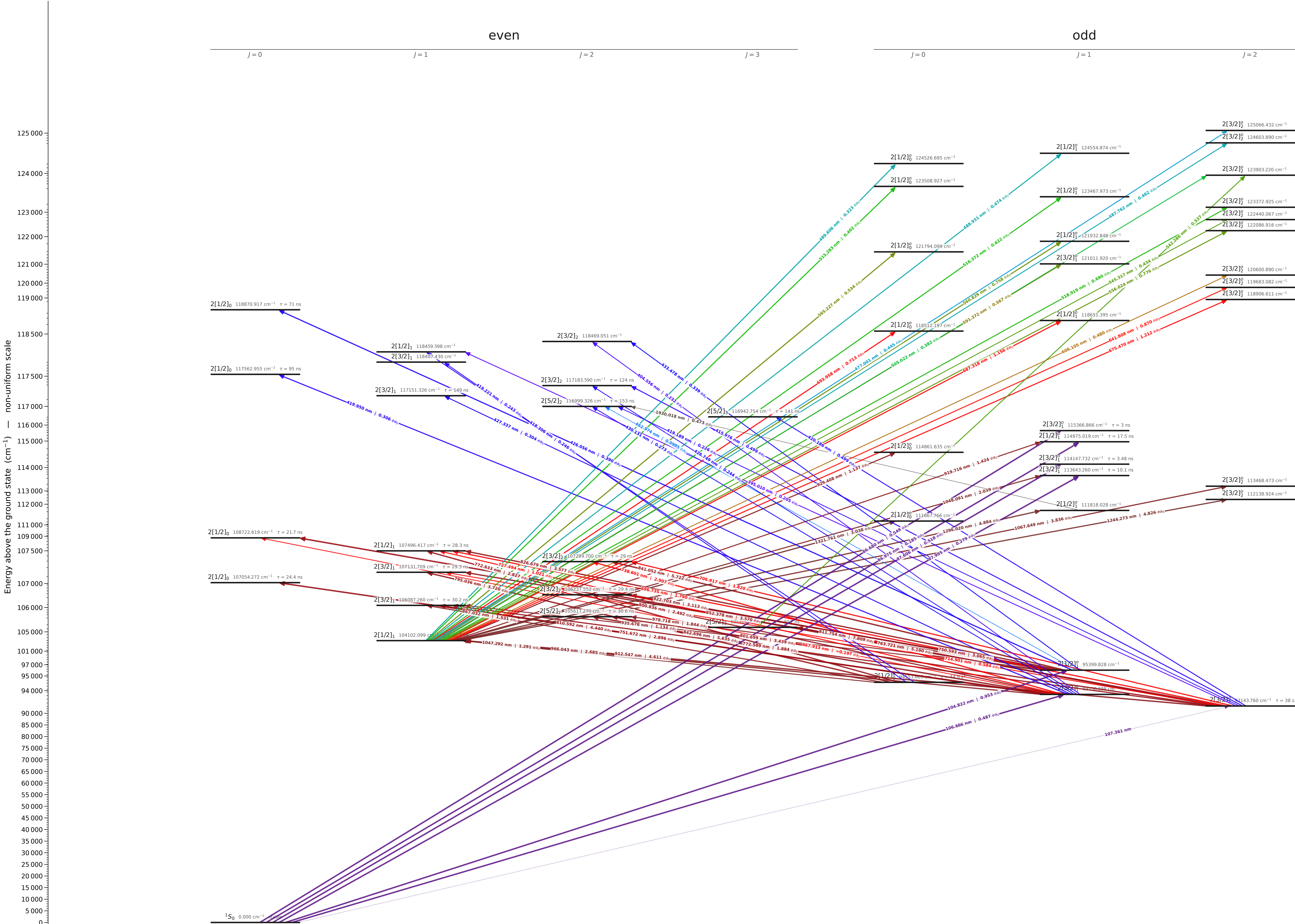
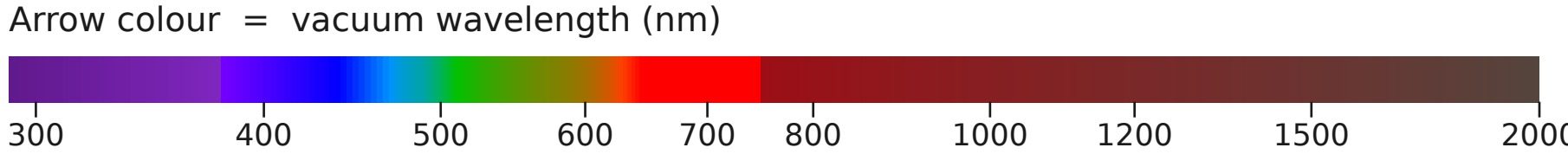


Argon

Ar (Ar I). Strongest classified transitions below 2 μm (NIST): 51 levels, 75 lines.

Ar⁺ ionisation limit 127109.84 cm⁻¹ ≈ 15.75961 eV (78.672 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| e r \| J' \rangle|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ e r \ J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ - 2[1/2] ^o ₁	104.8220	-	2860.01500(10)	0.953	5.320e+08	-	-	104.8 nm resonance line
¹ S ₀ - 2[3/2] ^o ₂	107.3609	-	2792.3797	-	2.630e-02	4.188e-09	99.9	107.4 nm M2 decay of the metastable 3P2 state
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.7207	763.5105	392.5420	5.190	2.450e+07	5.413	72	763.5 nm
2[3/2] ^o ₂ - 2[3/2] ^o ₁	800.8358	800.6156	374.3495	2.492	4.900e+06	5.413	14.4	800.8 nm optical pumping 1s4 -> 2p6 -> metastable 1s5
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.7542	811.5311	369.3143	7.809	3.300e+07	5.184	100	811.5 nm
2[1/2] ^o ₁ - 2[3/2] ^o ₁	922.7029	922.4498	324.9068	3.113	5.000e+06	5.413	14.7	922.7 nm optical pumping 1s2 -> 2p6 -> metastable 1s5

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

³⁹Ar hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ^o ₂	-285.36(13)	117.8(12)	F=3/2: +2631.35 F=5/2: +1865.36 F=7/2: +822.42 F=9/2: -468.01 F=11/2: -1968.07
2[3/2] ^o ₁	-334.0(20)	-24(3)	F=5/2: +1490.14 F=7/2: +351.14 F=9/2: -1175.00
2[1/2] ^o ₁	-712.0(10)	84(6)	F=5/2: +3249.00 F=7/2: +652.00 F=9/2: -2471.00
2[5/2] ^o ₃	-134.36(8)	113.1(14)	F=1/2: +1874.45 F=3/2: +1652.71 F=5/2: +1288.54 F=7/2: +790.00 F=9/2: +168.42 F=11/2: -561.68 F=13/2: -1382.51
2[3/2] ^o ₁	-163(31)	-63(11)	F=3/2: +1433.25 F=5/2: +1053.88 F=7/2: +507.00 F=9/2: -223.12 F=11/2: -1156.75

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
¹ S ₀ - 2[1/2] ^o ₁	104.822	40-38	550(10)
¹ S ₀ - 2[1/2] ^o ₁	104.822	40-36	1036(7)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	32-38	-882(4)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	33-38	-703.2(21)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	34-38	-552.6(7)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	35-38	-394.4(27)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	36-38	-265.4(4)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	37-38	-124.2(11)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	39-38	122(7)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	40-38	229.5(4)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	763.721	46-38	835.2(23)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	800.836	36-40	-640(10)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	800.836	38-40	-290(10)
2[3/2] ^o ₂ - 2[3/2] ^o ₁	800.836	39-40	-128(16)
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.754	36-38	-242.23(25)
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.754	40-38	207.9(9)
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.754	39-38	113.9(4)
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.754	37-38	-113.0(27)
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.754	36-40	-450.1(9)
2[3/2] ^o ₂ - 2[5/2] ^o ₃	811.754	39-40	-94.0(10)
2[1/2] ^o ₁ - 2[3/2] ^o ₁	922.703	36-40	-960(10)
2[1/2] ^o ₁ - 2[3/2] ^o ₁	922.703	38-40	-445(10)

Shift = ν(first isotope) - ν(second isotope).